

The empirical recurrence for OEIS A302419, recovered from its tail

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Abstract

OEIS A302419 records “Empirical recurrence of order 79” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 227$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 8$, so the recurrence is proved for every $n \geq 87$. Its last coefficient is $c_{79} = -4 \neq 0$, so no further tail satisfies anything shorter; any order-79 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302419 is “Number of $6 \times n$ 0..1 arrays with every element equal to 0, 1, 2 or 5 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

32, 857, 1431, 3548, 16123, 84581, 306064, 1137414, ...

The entry states:

Empirical recurrence of order 79 (see link above).

As of the “Last modified” line on the live entry (Jun 05 2026 14:32:42, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 227$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 227$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 79: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 8$ is the first whose minimal order is exactly the stated 79.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 454$ exact terms from $n = 8$ on, it returns order 79 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{79} c_i a(n-i),$$

c_1	7	c_{21}	−38546	c_{41}	530729	c_{61}	300
c_2	−17	c_{22}	40204	c_{42}	−402200	c_{62}	−24393
c_3	31	c_{23}	11158	c_{43}	−572594	c_{63}	28807
c_4	−39	c_{24}	−7934	c_{44}	257108	c_{64}	12116
c_5	−68	c_{25}	−7684	c_{45}	182904	c_{65}	−7360
c_6	276	c_{26}	−747	c_{46}	57011	c_{66}	28838
c_7	−419	c_{27}	33811	c_{47}	−196619	c_{67}	−2125
c_8	381	c_{28}	−138162	c_{48}	−63483	c_{68}	−5880
c_9	181	c_{29}	−103348	c_{49}	924632	c_{69}	11690
c_{10}	−1233	c_{30}	137681	c_{50}	−906010	c_{70}	−2312
c_{11}	1444	c_{31}	−72099	c_{51}	246158	c_{71}	−1968
c_{12}	1630	c_{32}	−17576	c_{52}	134590	c_{72}	2757
c_{13}	−5309	c_{33}	66458	c_{53}	−427786	c_{73}	−325
c_{14}	5331	c_{34}	496821	c_{54}	295084	c_{74}	42
c_{15}	−1289	c_{35}	−236390	c_{55}	−113500	c_{75}	88
c_{16}	1093	c_{36}	20451	c_{56}	72701	c_{76}	−147
c_{17}	−2874	c_{37}	393942	c_{57}	39591	c_{77}	98
c_{18}	−4618	c_{38}	24873	c_{58}	32087	c_{78}	18
c_{19}	5568	c_{39}	−601718	c_{59}	−30980	c_{79}	−4
c_{20}	7157	c_{40}	−117121	c_{60}	9031		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 87$, and it is the recurrence OEIS A302419 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 8$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{79} - c_1 x^{79-1} - \dots - c_{79}$. Its constant term is $-c_{79} = 4$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 79 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 79, so $\deg q = 79$, and it is monic. It holds from some index t on. From $\max(t, 8)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 79, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 79, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -4 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-79 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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